Liwei Hu

PhD Candidate in Applied Mathematics, University of Bologna

liwei.hu2@unibo.it | liweihu2.github.io

Research Interests

Computational inverse problems; structure-aware regularization; uncertainty quantification; numerical optimization; geophysical and Earth-system applications.

Education

PhD, Applied Mathematics — *Bologna, Italy*
University of Bologna — *2023–present*

- Advisor: Fabiana Zama
- Co-Advisors: Lisa Borgatti, Germana Landi, Davide Donati, Giacomo Titti
- Doctoral Program: EIT4SEMM

MS, Applied Mathematics — *Bologna, Italy*
University of Bologna — *2020–2023*

- Thesis Advisor: Fabiana Zama

BS, Mathematics — *Bologna, Italy*
University of Bologna — *2017–2020*

Research Experience

Case Western Reserve University — *Cleveland, OH*
Mentors: Erkki Somersalo, Daniela Calvetti — *Sep–Dec 2025*

- Role: Visiting PhD Researcher
- Topic: Bayesian scientific computing

University of California, Berkeley — *Berkeley, CA*
Mentor: Roland Bürgmann — *Apr–Jul 2025*

- Role: Visiting PhD Researcher
- Topic: Bayesian inverse problems for Earth surface dynamics

Publications

† Authors listed in alphabetical order

Published

†Borgatti L., Donati D., **Hu L.**, Landi G., Zama F. (2026). Estimating landslide thickness through domain-based regularization. *Journal of Computational and Applied Mathematics*, 476, 117073, pp. 1–15.

Titti G., **Hu L.**, Festi P., Elia L., Borgatti L., Lombardo L. (2025). An updated version of the SZ-plugin: From space to space–time data-driven modeling in QGIS. *International Journal of Applied Earth Observation and Geoinformation*, 142, 104679, pp. 1–13.

Spinelli G.V., Evangelista D., **Hu L.**, Zama F. (2025). Neural network-based inversion of NMR dispersion profiles for enhanced analysis of food systems. *Neural Computing & Applications*, 37, pp. 3957–3972.

†Borgatti L., Donati D., **Hu L.**, Landi G., Zama F. (2024). A regularization method for landslide thickness estimation. *Journal of Imaging*, 10(12): 314, pp. 1–19.

Submitted

Hu L., Calvetti D., Landi G., Somersalo E., Zama F. (2026). Bayesian variational approach to landslide thickness estimation. Submitted to *Inverse Problems*.

Manuscripts in Preparation

Hu L., Calvetti D., Landi G., Somersalo E., Zama F. (2026). Uncertainty quantification in landslide thickness estimation. Manuscript in preparation for *SIAM/ASA Journal on Uncertainty Quantification*.

Hu L., Bürgmann R., Xu Y., Hu X., Donati D., Landi G., Zama F., Borgatti L. (2026). Inferring landslide slip surface using kinematic elements. Manuscript in preparation for *Journal of Geophysical Research: Earth Surface*.

Presentations

* presenting author

Contributed Presentations

Hu L.*, Calvetti D., Landi G., Somersalo E., Zama F. (2026). *Regularization for landslide thickness estimation via finite element methods*. Oral presentation, SIAM Annual Meeting 2026, Cleveland, OH.

Hu L.*, Bürgmann R., Xu Y., Hu X., Donati D., Landi G., Zama F., Borgatti L. (2025). *Locating landslide slip surfaces using kinematic elements: A case study for Slumgullion landslide*. Geoburst oral presentation, AGU Annual Meeting 2025, New Orleans, LA.

Borgatti L., Donati D., Hu L.*, Landi G., Zama F. (2025). *A regularization method for landslide thickness estimation*. Departmental seminar, Department of Mathematics, University of Bologna, Bologna, Italy.

Hu L.*, Donati D., Landi G., Zama F., Borgatti L. (2024). *Development and application of an enhanced inversion algorithm for the reconstruction of the basal surface of landslides*. PhD Day oral presentation, Italian Geological Society-Italian Society of Mineralogy and Petrology 2024 Congress: Geology for sustainable management of our Planet, Bari, Italy.

Borgatti L., Donati D., Hu L.*, Landi G., Zama F. (2024). *A regularisation approach to landslide thickness estimation*. Oral presentation, Workshop on Advanced Numerical Analysis for Imaging: from theory to industry, Naples, Italy.

Poster Presentations

Hu L.*, Bürgmann R., Xu Y., Hu X., Donati D., Landi G., Zama F., Borgatti L. (2025). *Locating landslide slip surfaces using kinematic elements: A case study for Slumgullion landslide*. Poster presentation, AGU Annual Meeting 2025, New Orleans, LA.

Borgatti L., Donati D., Hu L.*, Landi G., Zama F. (2024). *An inverse problem from engineering geology: a regularisation approach for landslide thickness estimation*. Poster presentation, 4th IMA Conference on Inverse Problems from Theory to Application, Bath, UK.

Teaching Experience

Fall 2025	**Directed reading program in discrete inverse problems**, Graduate Mentor, Department of Mathematics, Applied Mathematics, and Statistics, Case Western Reserve University	*Cleveland, OH*
Spring 2025	**Numerical Differential Equations**, Teaching Assistant, Department of Mathematics, University of Bologna	*Bologna, Italy*
Fall 2024	**Numerical Methods**, Teaching Assistant, Department of Mathematics, University of Bologna	*Bologna, Italy*
Spring 2024	**Numerical Differential Equations**, Teaching Assistant, Department of Mathematics, University of Bologna	*Bologna, Italy*
Fall 2023	**Numerical Methods**, Teaching Assistant, Department of Mathematics, University of Bologna	*Bologna, Italy*

Awards, Fellowships, & Grants

2025	**Marco Polo Mobility Grant**, University of Bologna	*€ 8,100*
2023	**Emilia-Romagna Region Award**, Emilia-Romagna Region	*€ 15,000*
2023–2026	**PhD Fellowship**, University of Bologna	*€ 56,415 (3 years; fully funded)*